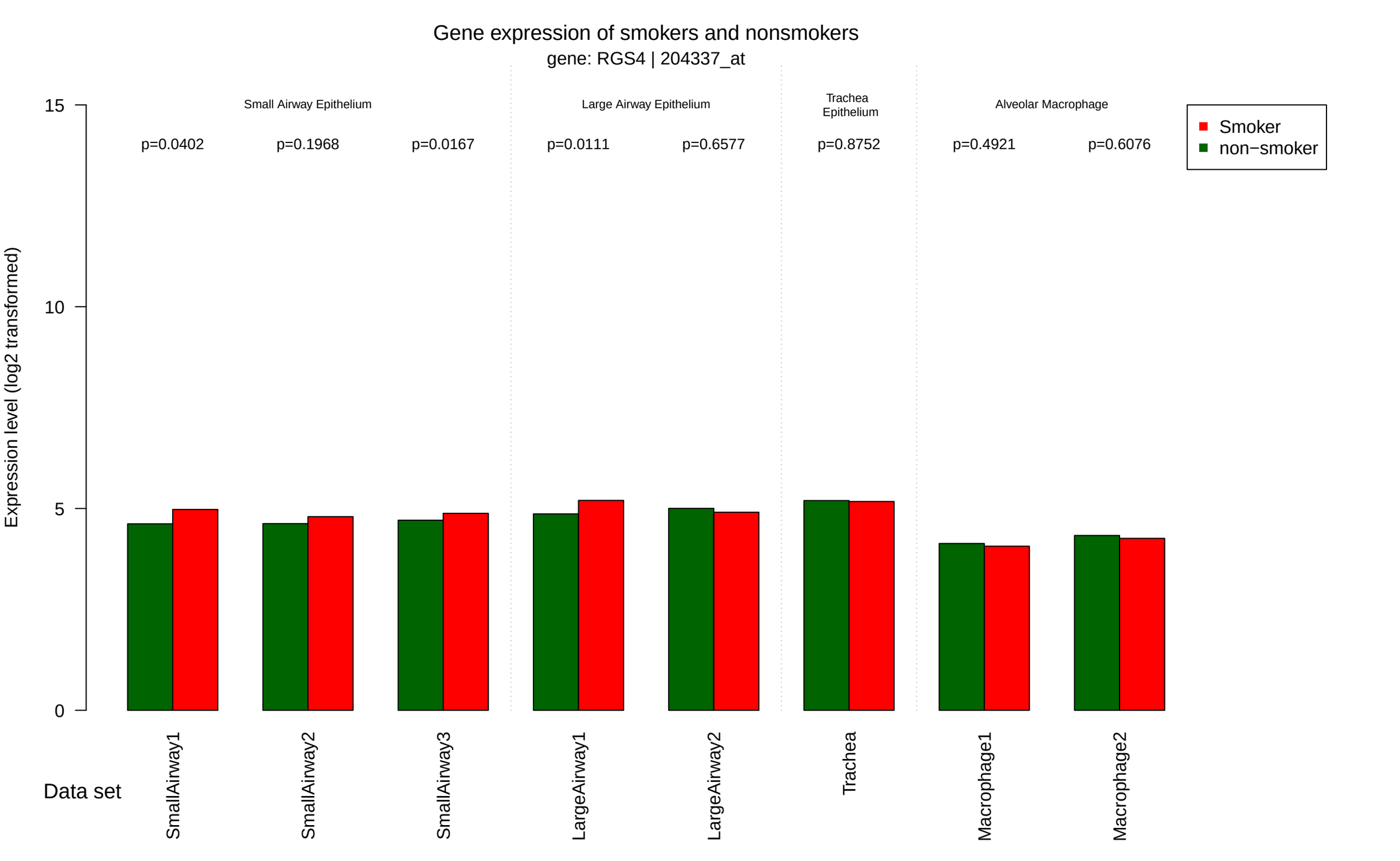
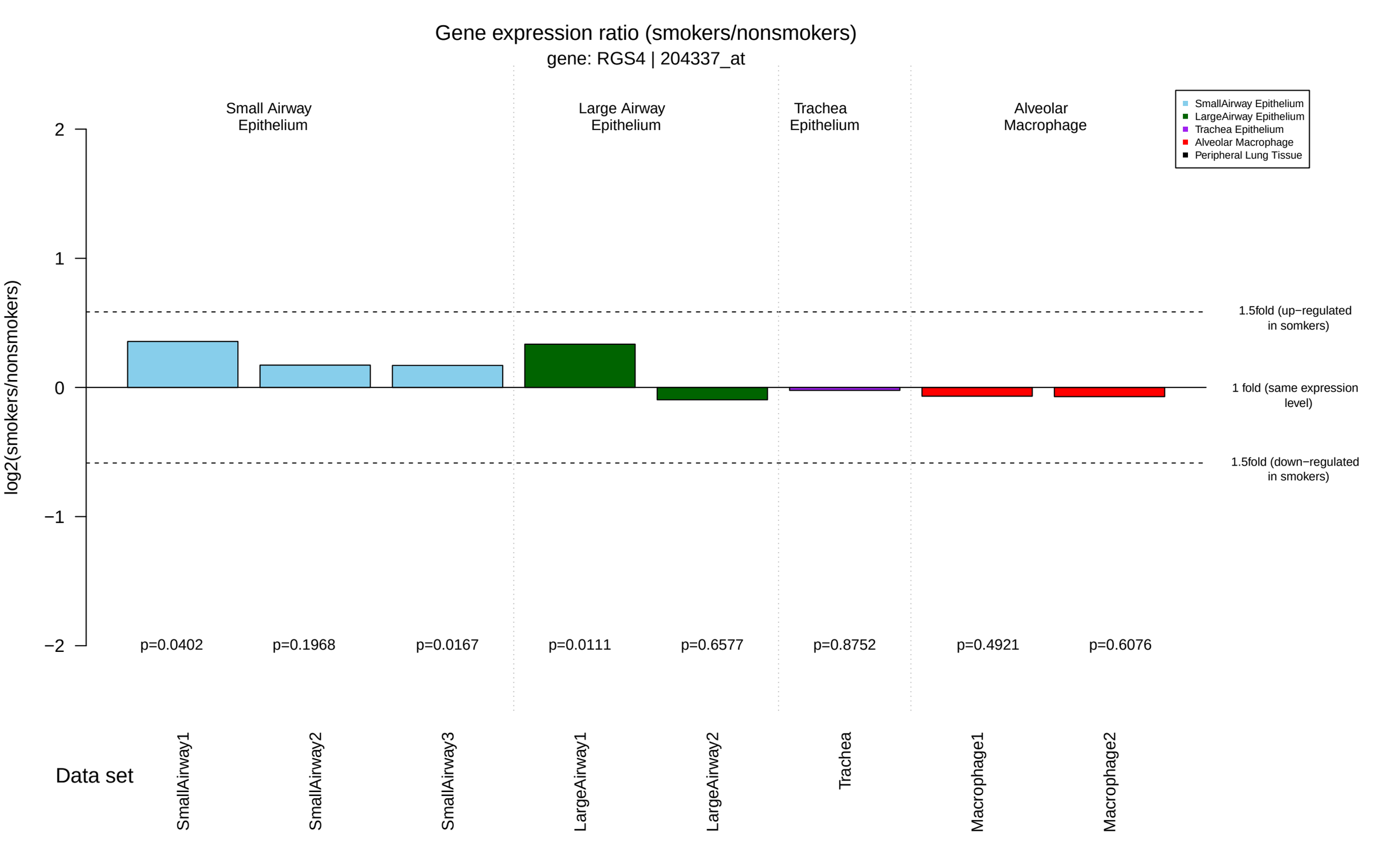
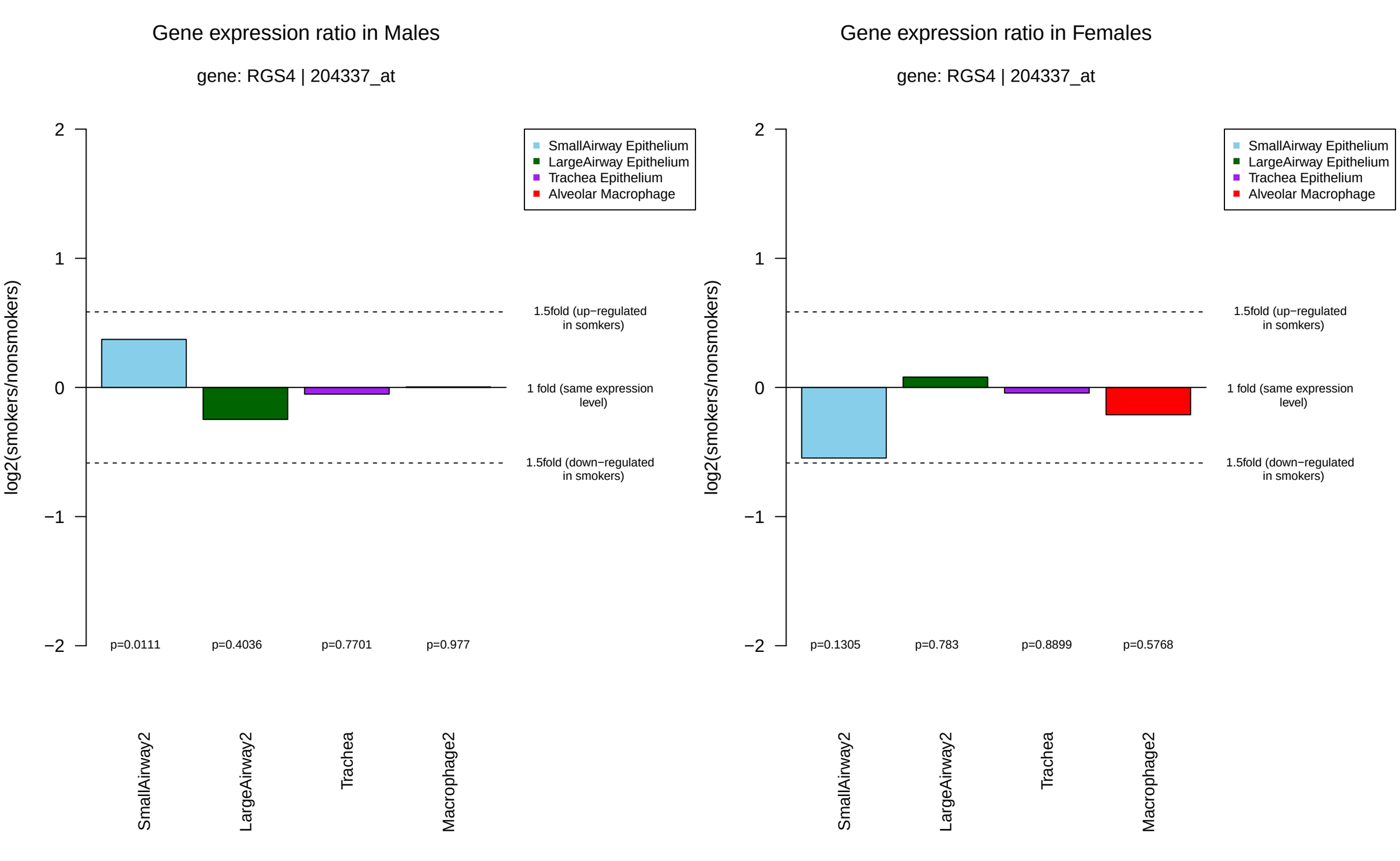
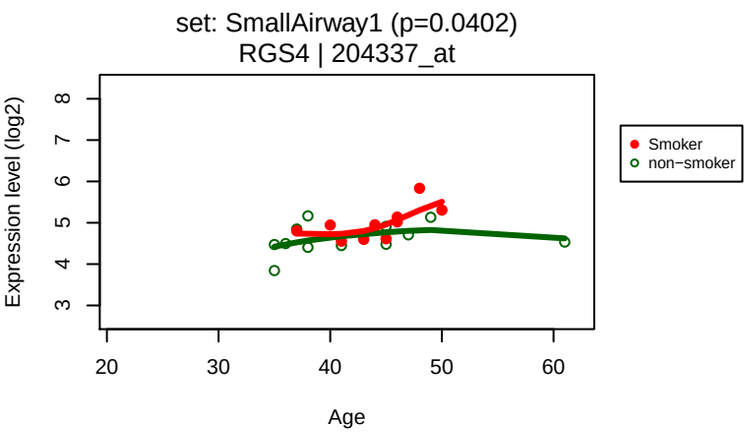
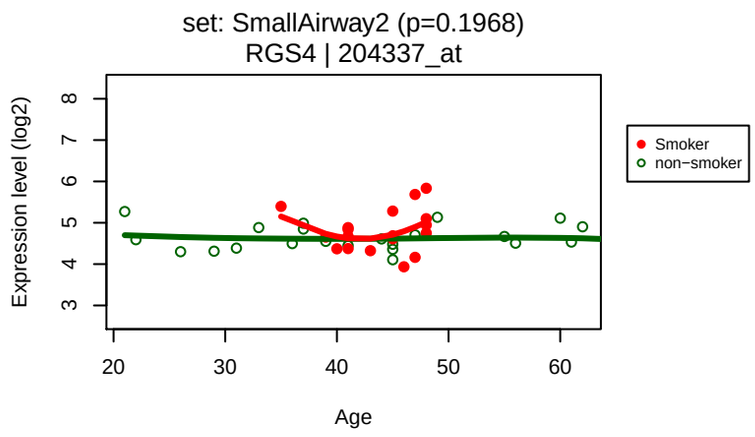
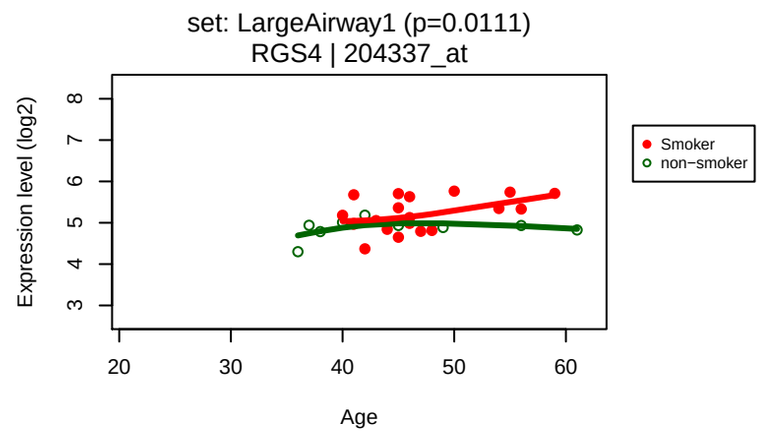

Gene expression of smokers and nonsmokers
gene: RGS4 | 204337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0402
p=0.1968
p=0.0167
p=0.0111
p=0.6577
p=0.8752
p=0.4921
p=0.6076
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

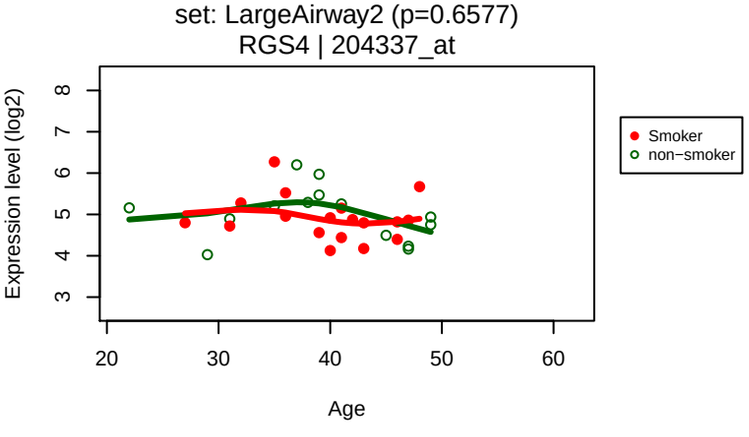

Gene expression ratio (smokers/nonsmokers)
gene: RGS4 | 204337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0402
p=0.1968
p=0.0167
p=0.0111
p=0.6577
p=0.8752
p=0.4921
p=0.6076
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

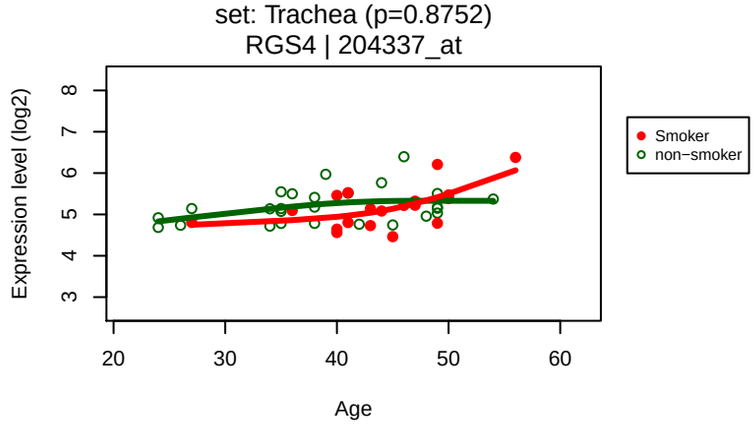

Gene expression ratio in Males
gene: RGS4 | 204337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0111
p=0.4036
p=0.7701
p=0.977
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS4 | 204337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1305
p=0.783
p=0.8899
p=0.5768
SmallAirway2
LargeAirway2
Trachea
Macrophage2

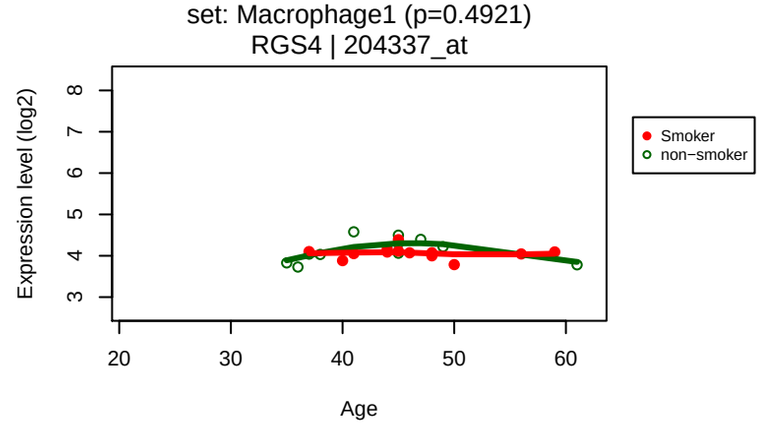

set: SmallAirway1 (p=0.0402)
RGS4 | 204337_at
Expression level (log2)
Age
Smoker
non−smoker

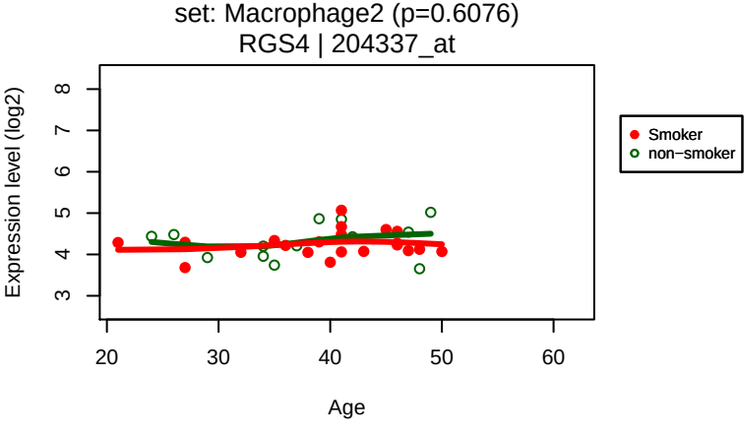

set: SmallAirway2 (p=0.1968)
RGS4 | 204337_at
Expression level (log2)
Age
Smoker
non−smoker

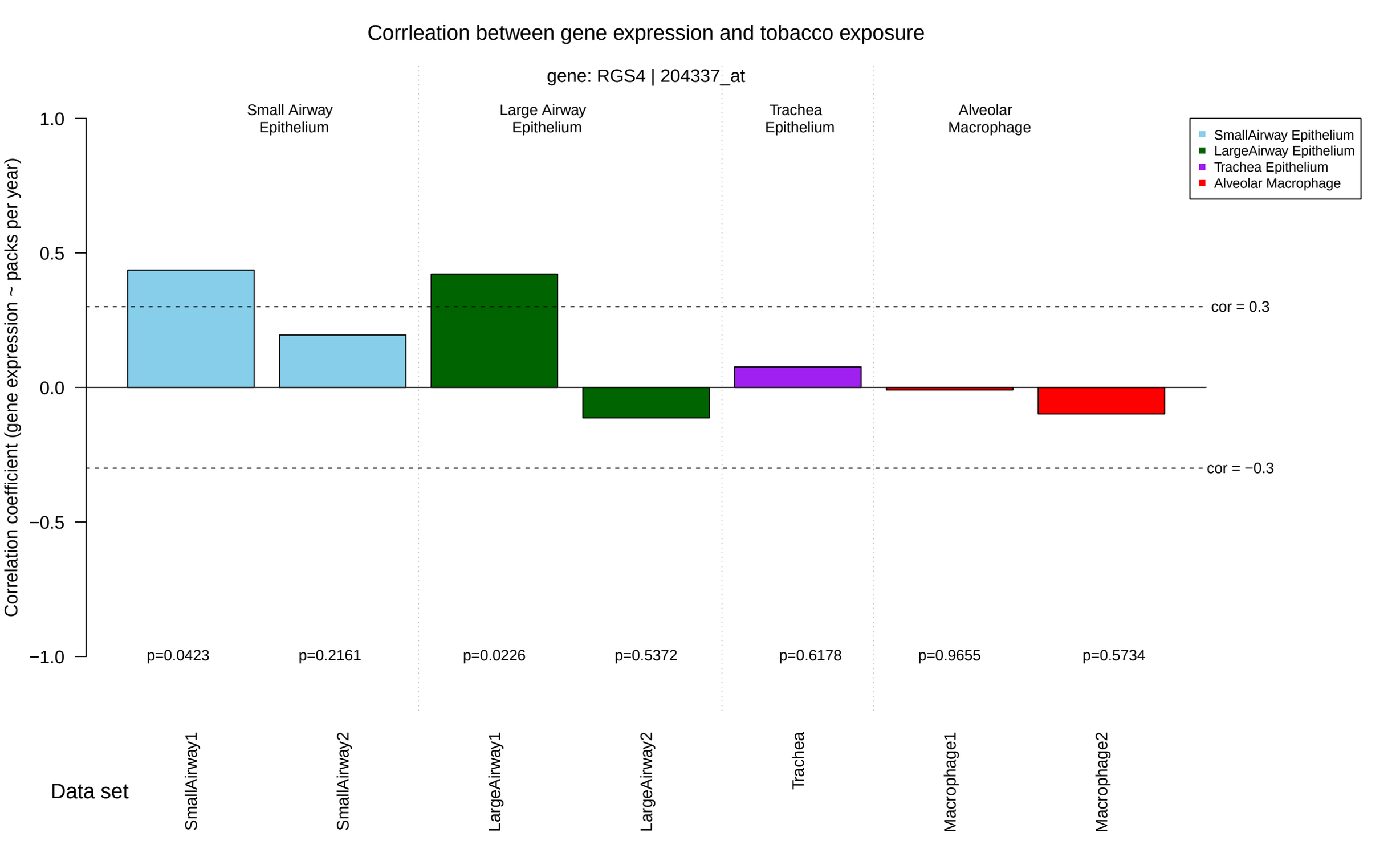

set: LargeAirway1 (p=0.0111)
RGS4 | 204337_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6577)
RGS4 | 204337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8752)
RGS4 | 204337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4921)
RGS4 | 204337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6076)
RGS4 | 204337_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS4 | 204337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0423
p=0.2161
p=0.0226
p=0.5372
p=0.6178
p=0.9655
p=0.5734
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2